

TASK 4:

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 12 15:59

affan-afk@affan-afk-VirtualBox: ~/Desktop

affan-afk@affan-afk-VirtualBox:~/Desktop$ nano isPrime.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ chmod +x isPrime.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat isPrime.sh
read -p "Enter a Number: " number
isPrime=1
if [[ $number -lt 2 ]]; then
    isPrime=0
else
    for i in $(seq 2 $(number-1)); do
        if [[ $number%i -eq 0 ]]; then
            isPrime=0
        fi
    done
fi
if [[ $isPrime -eq 1 ]]; then
    echo $number " is prime"
else
    echo $number " is not prime!"
fi
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./isPrime.sh
Enter a Number: 1
1 is not prime!
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./isPrime.sh
Enter a Number: 2
2 is prime
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./isPrime.sh
Enter a Number: 3
3 is prime
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./isPrime.sh
Enter a Number: 4
4 is not prime!
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./isPrime.sh
Enter a Number: 5
5 is prime
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./isPrime.sh
```

TASK 5: (tried using arrays but got stuck)

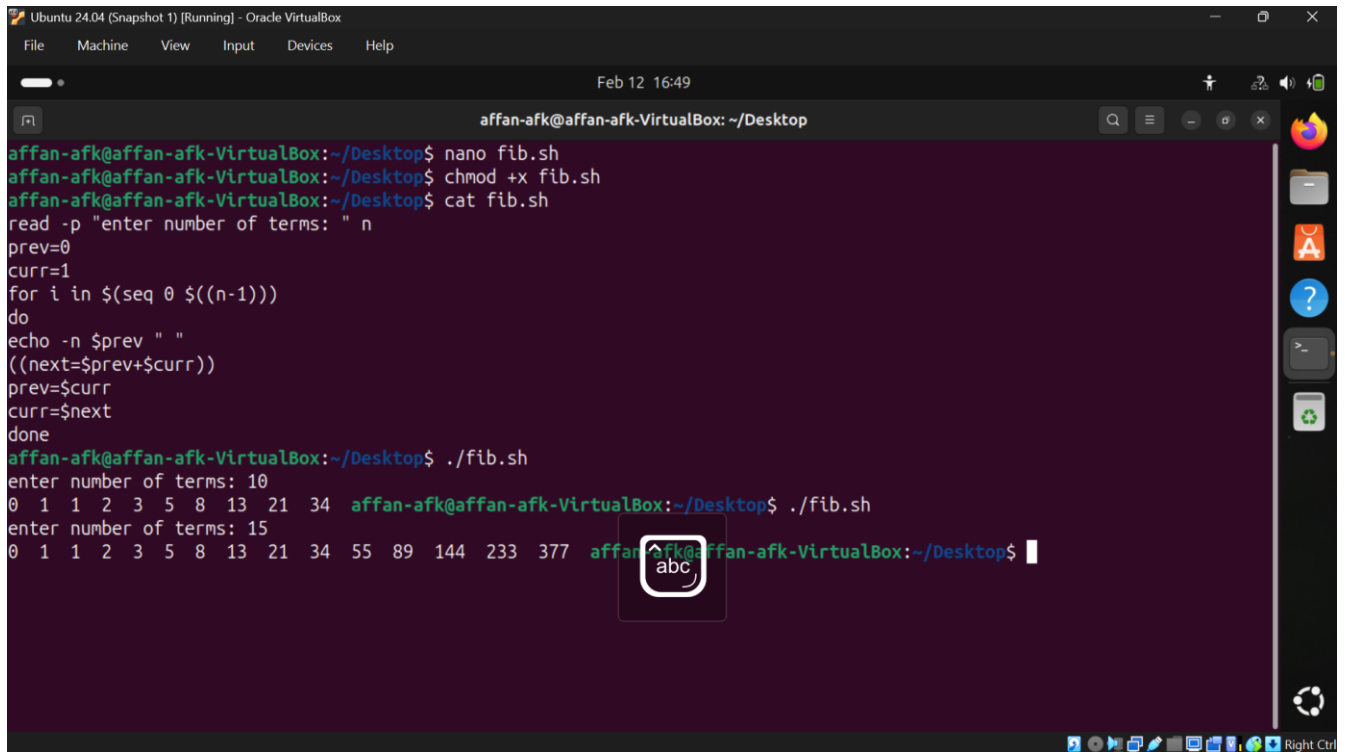
```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
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Feb 12 16:34

affan-afk@affan-afk-VirtualBox: ~/Desktop

affan-afk@affan-afk-VirtualBox:~/Desktop$ nano MinMax.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat MinMax.sh
read -p "Enter 5 numbers separated by space: " a b c d e
max=$a
min=$a
for i in $a $b $c $d $e
do
    if [[ $i -gt $max ]]; then
        max=$i
    fi
    if [[ $i -lt $min ]]; then
        min=$i
    fi
done
echo "min: " $min
echo "max: " $max
affan-afk@affan-afk-VirtualBox:~/Desktop$ chmod +x MinMax.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./MinMax.sh
Enter 5 numbers separated by space: 34 21 333 1 3
min: 1
max: 333
affan-afk@affan-afk-VirtualBox:~/Desktop$
```

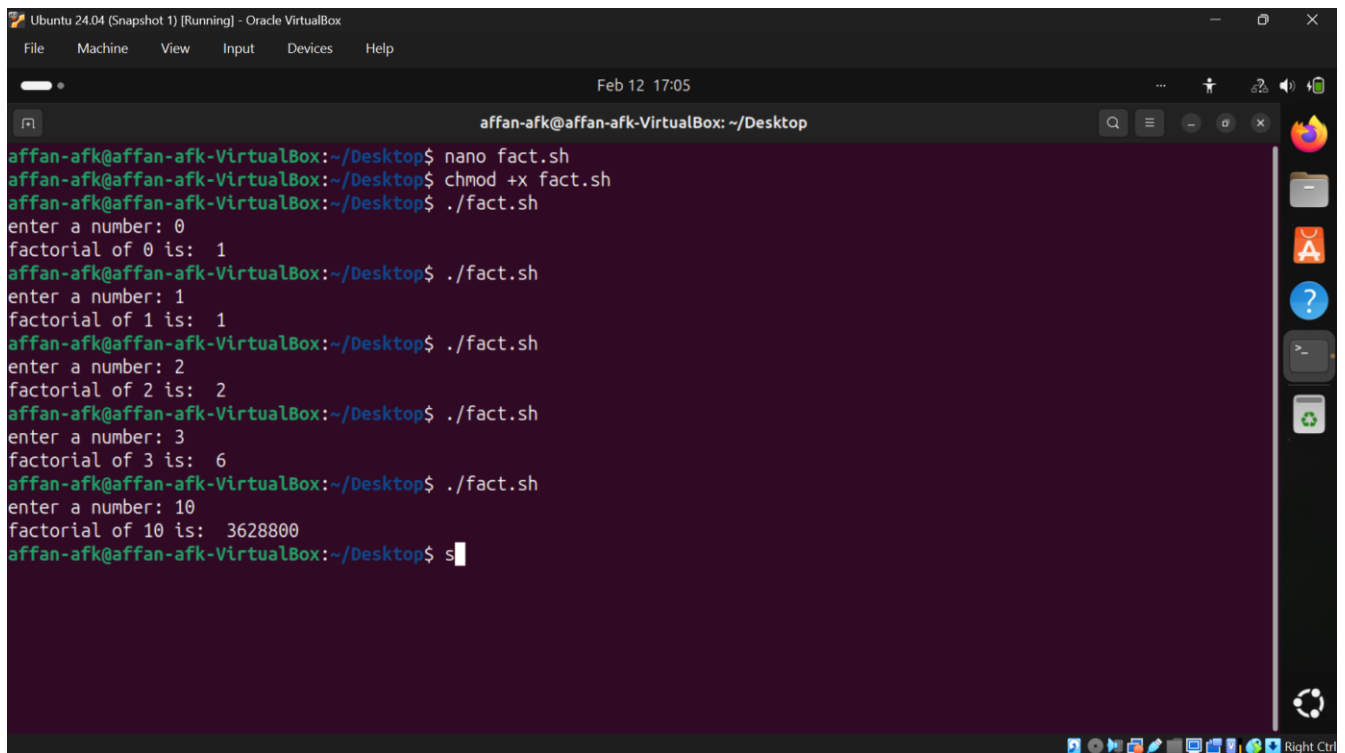
TASK6:



The screenshot shows a terminal window titled "affan-afk@affan-afk-VirtualBox: ~/Desktop". The user has created a script named "fib.sh" and made it executable. The script prompts the user to "enter number of terms: " and then prints the first 'n' terms of the Fibonacci sequence. The user has run the script twice: first with 10 terms, resulting in the sequence 0 1 1 2 3 5 8 13 21 34, and then with 15 terms, resulting in the sequence 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377. A small "abc" logo is visible in the center of the terminal window.

```
affan-afk@affan-afk-VirtualBox: ~/Desktop
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano fib.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ chmod +x fib.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat fib.sh
read -p "enter number of terms: " n
prev=0
curr=1
for i in $(seq 0 $((n-1)))
do
echo -n $prev " "
((next=$prev+$curr))
prev=$curr
curr=$next
done
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fib.sh
enter number of terms: 10
0 1 1 2 3 5 8 13 21 34
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fib.sh
enter number of terms: 15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
affan-afk@affan-afk-VirtualBox:~/Desktop$
```

TASK 7:



The screenshot shows a terminal window titled "affan-afk@affan-afk-VirtualBox: ~/Desktop". The user has created a script named "fact.sh" and made it executable. The script prompts the user to "enter a number:" and then prints the factorial of that number. The user has run the script five times with inputs 0, 1, 2, 3, and 10, resulting in outputs 1, 1, 2, 6, and 3628800 respectively. The user has also typed the letter 's' on the last line.

```
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano fact.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ chmod +x fact.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fact.sh
enter a number: 0
factorial of 0 is: 1
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fact.sh
enter a number: 1
factorial of 1 is: 1
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fact.sh
enter a number: 2
factorial of 2 is: 2
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fact.sh
enter a number: 3
factorial of 3 is: 6
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./fact.sh
enter a number: 10
factorial of 10 is: 3628800
affan-afk@affan-afk-VirtualBox:~/Desktop$ s
```

TASK 8:

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
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Feb 12 23:55

affan-afk@affan-afk-VirtualBox: ~/Desktop/shelldir

affan-afk@affan-afk-VirtualBox:~/Desktop$ nano task9.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat task9.sh
#!/bin/bash

src="/home/affan-afk/Desktop"
dst1="/home/affan-afk/Desktop/shelldir"
dst2="/home/affan-afk/Desktop/cdir"
dst3="/home/affan-afk/Desktop/jpgdir"

if [ ! -d "$dst1" ]; then
    mkdir -p "$dst1"
fi
if [ ! -d "$dst2" ]; then
    mkdir -p "$dst2"
fi
if [ ! -d "$dst3" ]; then
    mkdir -p "$dst3"
fi
for file in "$src"/*.sh; do
    mv "$file" "$dst1"
done
for file in "$src"/*.c; do
    mv "$file" "$dst2"
done
for file in "$src"/*.jpg; do
    mv "$file" "$dst3"
done
affan-afk@affan-afk-VirtualBox:~/Desktop$ chmod +x task9.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ ls
cdir          'Screenshot From 2026-02-12 23-49-56 (Copy 2).jpg'
event.c       'Screenshot From 2026-02-12 23-49-56 (Copy 3).jpg'
event_reservation.h 'Screenshot From 2026-02-12 23-49-56 (Copy 4).jpg'
events        'Screenshot From 2026-02-12 23-49-56 (Copy).jpg'
fact.sh       'Screenshot From 2026-02-12 23-49-56.jpg'
fib.sh        shelldir
grade_calculator simple sort
oracle_calculator
```

```
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affan-afk@affan-afk-VirtualBox: ~/Desktop/shelldir

mv "$file" "$dst3"
done
affan-afk@affan-afk-VirtualBox:~/Desktop$ chmod +x task9.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ ls
cdir          'Screenshot From 2026-02-12 23-49-56 (Copy 2).jpg'
event.c       'Screenshot From 2026-02-12 23-49-56 (Copy 3).jpg'
event_reservation.h 'Screenshot From 2026-02-12 23-49-56 (Copy 4).jpg'
events        'Screenshot From 2026-02-12 23-49-56 (Copy).jpg'
fact.sh       'Screenshot From 2026-02-12 23-49-56.jpg'
fib.sh        shelldir
grade_calculator simple sort
grade_calculator.c simple_sort.c
isPrime.sh    sum_average
main_event.c  sum_average.c
makefile      task9.sh
MinMax.sh     task9.sh.save
missing_sequence test.sh
missing_sequence.c

affan-afk@affan-afk-VirtualBox:~/Desktop$ ./task9.h
bash: ./task9.h: No such file or directory
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./task9.sh
affan-afk@affan-afk-VirtualBox:~/Desktop$ ls
cdir          grade_calculator missing_sequence sum_average
event_reservation.h jpgdir          shelldir      task9.sh.save
events        makefile          simple_sort
affan-afk@affan-afk-VirtualBox:~/Desktop$ cd cdir
affan-afk@affan-afk-VirtualBox:~/Desktop/cdir$ ls
event.c      main_event.c  simple_sort.c
grade_calculator.c missing_sequence.c sum_average.c
affan-afk@affan-afk-VirtualBox:~/Desktop/cdir$ cd ..
affan-afk@affan-afk-VirtualBox:~/Desktop$ cd jpgdir
affan-afk@affan-afk-VirtualBox:~/Desktop/jpgdir$ ls
'Screenshot From 2026-02-12 23-49-56 (Copy 2).jpg'
'Screenshot From 2026-02-12 23-49-56 (Copy 3).jpg'
'Screenshot From 2026-02-12 23-49-56 (Copy 4).jpg'
'Screenshot From 2026-02-12 23-49-56 (Copy).jpg'
affan-afk@affan-afk-VirtualBox:~/Desktop/jpgdir$ cd ..
affan-afk@affan-afk-VirtualBox:~/Desktop$ cd shelldir
affan-afk@affan-afk-VirtualBox:~/Desktop/shelldir$ ls
fact.sh  fib.sh  isPrime.sh  MinMax.sh  task9.sh  test.sh
affan-afk@affan-afk-VirtualBox:~/Desktop/shelldir$
```

TASK 9: